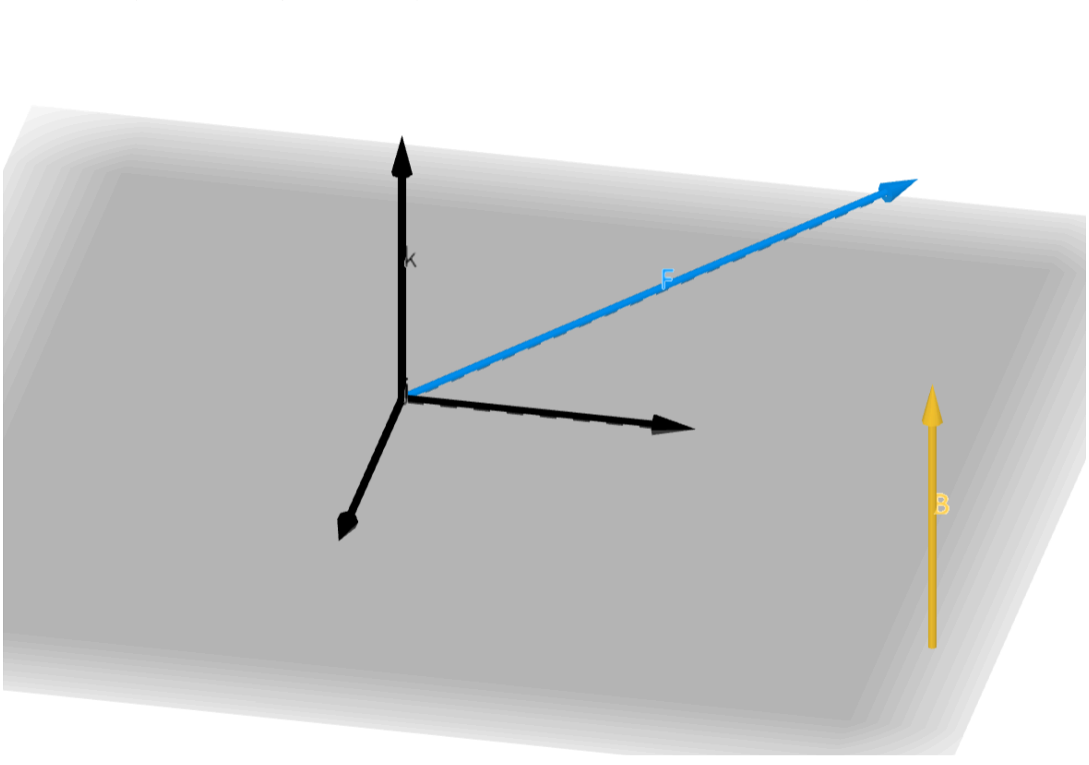
复合场（磁场与电场）中带电粒子的运动



设空间中存在$\overrightarrow{B}=B_0\hat{k}$ 和 $\overrightarrow{F}$
带正电粒子，初速度为 $\overrightarrow{v_0}$, 电荷量为q
$\overrightarrow{F}$ 大小为F_0

$$\overrightarrow{F}=F_x\hat{i}+F_y\hat{j}+F_z\hat{k}$$

$$\overrightarrow{v_0}=v_{0x}\hat{i}+v_{0y}\hat{j}+v_{0z}\hat{k}$$

由牛顿第二定律

$$q(\overrightarrow{v}\times\overrightarrow{B})+\overrightarrow{F}=m\overrightarrow{a}$$

其中

$$\begin{aligned}\overrightarrow{v}\times\overrightarrow{B}=&(v_x\hat{i}+v_y\hat{j}+v_z\hat{k})\times B_0\hat{k}\\=&B_0(v_x\hat{i}\times\hat{k}+v_y\hat{j}\times\hat{k}+v_z\hat{k}\times\hat{k})\\=&B_0(-v_x\hat{j}+v_y\hat{i})\\=&-v_xB_0\hat{j}+v_yB_0\hat{i}\end{aligned}$$

$$-qv_xB_0\hat{j}+qv_yB_0\hat{i}+F_x\hat{i}+F_y\hat{j}+F_z\hat{k}=ma_x\hat{i}+ma_y\hat{j}+ma_z\hat{k}$$

$$\begin{cases}qv_yB_0+F_x=ma_x(1)\\-qv_xB_0+F_y=ma_y(2)\\F_z=ma_z(3)\end{cases}$$

$\frac{(1)}{(2)}$ 得

$$\frac{v_y+\frac{F_x}{qB_0}}{-v_x+\frac{F_y}{qB_0}}=\frac{a_x}{a_y}=\frac{a_x\Delta x}{a_y\Delta y}=\frac{\Delta v_x}{\Delta v_y}$$

$$v_y\Delta v_y+\frac{F_x}{qB_0}\Delta v_y+v_x\Delta x-\frac{F_y}{qB_0}\Delta x=0$$

积分

$$\frac{1}{2}v_y^2+\frac{F_x}{qB_0}v_y+\frac{1}{2}v_x^2-\frac{F_x}{qB_0}v_x=\frac{1}{2}(v_{0x}^2+v_{0y}^2)+\frac{F_x}{qB_0}(v_{0y}-v_{0x})$$

$$(v_y+\frac{F_x}{qB_0})^2+(v_x-\frac{F_y}{qB_0})^2=(v_{0y}+\frac{F_x}{qB_0})^2+(v_{0x}-\frac{F_y}{qB_0})^2$$

设

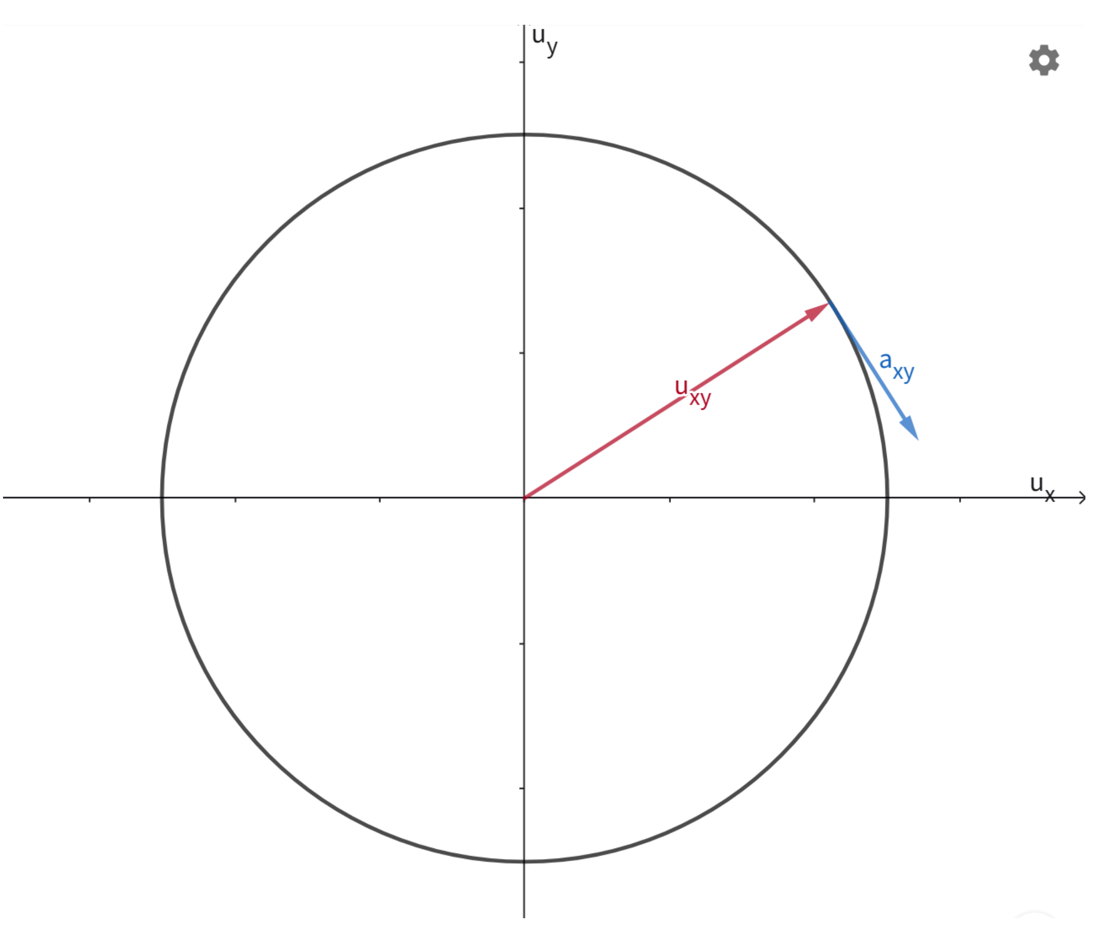
$$\overrightarrow{u}=(v_x-\frac{F_y}{qB_0})\hat{i}+(v_y+\frac{F_x}{qB_0})\hat{j}+v_z\hat{k}$$

$$\overrightarrow{u_0}=(v_{0x}-\frac{F_y}{qB_0})\hat{i}+(v_{0y}+\frac{F_x}{qB_0})\hat{j}+v_{0z}\hat{k}$$

则

$$u_x^2+u_y^2=u_{0x}^2+u_{0y}^2\text{即}\mid\overrightarrow{u_{xy}}\mid=\sqrt{u_{0x}^2+u_{0y}^2}$$

下面先只考虑x y方向



$$\overrightarrow{u_{xy}}=\left(v_x-\frac{F_y}{qB_0}\right)\hat{i}+\left(v_y+\frac{F_x}{qB_0}\right)\hat{j}$$

由(1)(2)得 $\frac{d\overrightarrow{u_{xy}}}{dt}=\overrightarrow{a_{xy}}=\left(\frac{qv_yB_0}{m}+\frac{F_x}{m}\right)\hat{i}+\left(\frac{-qv_xB_0}{m}+\frac{F_y}{m}\right)\hat{j}$

得

$$\overrightarrow{u_{xy}}\cdot\overrightarrow{a_{xy}}=0,\text{即}\overrightarrow{u_{xy}}\perp\frac{d\overrightarrow{u_{xy}}}{dt}$$

$$\frac{qB_0}{m}\mid\overrightarrow{u_{xy}}\mid=\mid\frac{d\overrightarrow{u_{xy}}}{dt}\mid$$

$$T=2\pi\frac{\mid\overrightarrow{u_{xy}}\mid}{\mid\frac{d\overrightarrow{u_{xy}}}{dt}\mid}=2\pi\frac{\mid\overrightarrow{u_{xy}}\mid}{\frac{qB_0}{m}\mid\overrightarrow{u_{xy}}\mid}=\frac{2\pi m}{qB_0}$$

$$\omega=\frac{2\pi}{T}$$

$$\begin{cases}u_x=\sqrt{u_{0x}^2+u_{0y}^2}\cos(\omega t+\varphi)\\u_y=\sqrt{u_{0x}^2+u_{0y}^2}\sin(\omega t+\varphi)\end{cases}$$

其中

$$\begin{cases}u_{x0}=\sqrt{u_{0x}^2+u_{0y}^2}\cos(\varphi)\\u_{y0}=\sqrt{u_{0x}^2+u_{0y}^2}\sin(\varphi)\end{cases}$$

$$\begin{cases}\sin\varphi=\frac{v_{0y}+\frac{F_x}{qB_0}}{\sqrt{(v_{0x}-\frac{F_y}{qB_0})^2+(v_{0y}+\frac{F_x}{qB_0})^2}}\\\cos\varphi=\frac{v_{0x}-\frac{F_y}{qB_0}}{\sqrt{(v_{0x}-\frac{F_y}{qB_0})^2+(v_{0y}+\frac{F_x}{qB_0})^2}}\end{cases}$$

$$\begin{cases}v_x=\frac{F_y}{qB_0}+\sqrt{u_{0x}^2+u_{0y}^2}\cos(\omega t+\varphi)\\v_y=-\frac{F_x}{qB_0}+\sqrt{u_{0x}^2+u_{0y}^2}\sin(\omega t+\varphi)\end{cases}$$

积分

$$\begin{cases}x=\frac{F_y}{qB_0}t+\frac{\sqrt{u_{0x}^2+u_{0y}^2}}{\omega}\sin(\omega t+\varphi)-\frac{\sqrt{u_{0x}^2+u_{0y}^2}}{\omega}\sin(\varphi)\\y=-\frac{F_x}{qB_0}t-\frac{\sqrt{u_{0x}^2+u_{0y}^2}}{\omega}\cos(\omega t+\varphi)+\frac{\sqrt{u_{0x}^2+u_{0y}^2}}{\omega}\cos(\varphi)\end{cases}$$

$$\begin{cases}x=\frac{F_y}{qB_0}t+\frac{m\sqrt{u_{0x}^2+u_{0y}^2}}{qB_0}\sin(\omega t+\varphi)-\frac{mu_{0y}}{qB_0}\\y=-\frac{F_x}{qB_0}t-\frac{m\sqrt{u_{0x}^2+u_{0y}^2}}{qB_0}\cos(\omega t+\varphi)+\frac{mu_{0x}}{qB_0}\end{cases}$$

下面考虑z方向

$$F_z=ma_z$$

$$z=v_{0z}t+\frac{1}{2}\left(\frac{F_z}{m}\right)t^2$$

结果

$$\begin{cases}x=\frac{F_y}{qB_0}t+\frac{m\sqrt{u_{0x}^2+u_{0y}^2}}{qB_0}\sin(\omega t+\varphi)-\frac{mu_{0y}}{qB_0}\\y=-\frac{F_x}{qB_0}t-\frac{m\sqrt{u_{0x}^2+u_{0y}^2}}{qB_0}\cos(\omega t+\varphi)+\frac{mu_{0x}}{qB_0}\\z=v_{0z}t+\frac{1}{2}\left(\frac{F_z}{m}\right)t^2\end{cases}$$

其中

$$\begin{cases}\sin\varphi=\frac{v_{0y}+\frac{F_x}{qB_0}}{\sqrt{(v_{0x}-\frac{F_y}{qB_0})^2+(v_{0y}+\frac{F_x}{qB_0})^2}}\\\cos\varphi=\frac{v_{0x}-\frac{F_y}{qB_0}}{\sqrt{(v_{0x}-\frac{F_y}{qB_0})^2+(v_{0y}+\frac{F_x}{qB_0})^2}}\end{cases}$$